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Roll No - 37

Assignment No 6

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In [1]:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classification_report, accuracy_score, precision_score, recall
from sklearn.preprocessing import LabelEncoder
```

In [2]:

```
data = pd.read_csv('Iris.csv')
data.head(5)
```

Out[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [3]:

```
data.describe(include = 'all')
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
count	150.000000	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	NaN	Iris-setosa
freq	NaN	NaN	NaN	NaN	NaN	50

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Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
count	150.000000	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	NaN	Iris-setosa
freq	NaN	NaN	NaN	NaN	NaN	50
mean	75.500000	5.843333	3.054000	3.758667	1.198667	NaN
std	43.445368	0.828066	0.433594	1.764420	0.763161	NaN
min	1.000000	4.300000	2.000000	1.000000	0.100000	NaN
25%	38.250000	5.100000	2.800000	1.600000	0.300000	NaN
50%	75.500000	5.800000	3.000000	4.350000	1.300000	NaN
75%	112.750000	6.400000	3.300000	5.100000	1.800000	NaN
max	150.000000	7.900000	4.400000	6.900000	2.500000	NaN

In [4]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Id               150 non-null   int64
1   SepalLengthCm    150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   Species          150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

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```
In [5]: print(data.shape)
data['Species'].unique()

(150, 6)

Out[5]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)

In [6]: data.isnull().sum()

Out[6]: Id                0
SepalLengthCm          0
SepalWidthCm           0
PetalLengthCm          0
PetalWidthCm           0
Species                0
dtype: int64

In [7]: x = data.iloc[:,1:5]
y = data.iloc[:,5:]

In [8]: encode = LabelEncoder()
y = encode.fit_transform(y)

C:\Users\khand\anaconda3\Lib\site-packages\sklearn\preprocessing\_label.py:114: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)

In [9]: x_train,x_test,y_train,y_test = train_test_split(x,y,test_size = 0.3,random_state = 0)

In [10]: naive_bayes = GaussianNB()
naive_bayes.fit(x_train,y_train)
pred = naive_bayes.predict(x_test)

In [11]: pred
```

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```
In [11]: pred

Out[11]: array([2, 1, 0, 2, 0, 2, 0, 1, 1, 1, 2, 1, 1, 1, 0, 1, 1, 0, 0, 2, 1,
               0, 0, 2, 0, 0, 1, 1, 0, 2, 1, 0, 2, 2, 1, 0, 1, 1, 1, 2, 0, 2, 0,
               0])

In [12]: y_test

Out[12]: array([2, 1, 0, 2, 0, 2, 0, 1, 1, 1, 2, 1, 1, 1, 0, 1, 1, 0, 0, 2, 1,
               0, 0, 2, 0, 0, 1, 1, 0, 2, 1, 0, 2, 2, 1, 0, 1, 1, 1, 2, 0, 2, 0,
               0])

In [13]: matrix = confusion_matrix(y_test,pred,labels = naive_bayes.classes_)
print(matrix)

tp, fn, fp, tn = confusion_matrix(y_test,pred,labels=[1,0]).reshape(-1)

[[16  0  0]
 [ 0 18  0]
 [ 0  0 11]]

In [14]: conf_matrix = ConfusionMatrixDisplay(confusion_matrix=matrix,display_labels=naive_bayes.classes_)
conf_matrix.plot(cmap=plt.cm.YlGn)
plt.show()
```



